

System Design Report

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Chapter 1: Requirements and Mechanics

1.1: System Requirements

One of the most important things we need to be aware of, is the requirements for the Micromouse, this includes; speed, operating environment, estimated mass, reliability etc. list from template.

The software development for the Micromouse will be done via Keil Studio, a cloud-based development environment provided by Arm. Keil Studio supports C++ programming and is compatible with the OM11043 LPC1768 MBED Board. This platform is designed for compiling, debugging and uploading directly to the microcontroller making it ideal for this project.

Table 1: Estimated Mass

Component	Estimated Mass
LPC1768 MCU (on MBED board)	30g
L297 Stepper Motor Controller	3g (± 0.5 g)
Y129 Unipolar Stepper Motor	220g
ULN2003 Motor Driver	~ 12 g
OP165D IR Emitter	~ 1 g
IS471 IR Receiver	~ 1 g
Chassis	50g
Battery Holder	75g
AA Batteries x 8	184g
Total	576g

The estimated speed of the Micromouse is up to 7 m/s (Danie Conradie, "The Fascinating Evolution of Micro mouse," *Hackaday*, May 29, 2023) [1]

The most important requirement for a Micromouse is its reliability as multiple things can go wrong during a run. The Micromouse must detect the walls using the IR sensors accurately to know when to turn or remain centred. We will need noise suppression for the IR sensors to prevent any misreadings, this will allow precise cell to cell movement through a more precise wall detection. Our mouse will be powered by 8x AA batteries to reach the 12V required by the stepper motors. For the rest of the components which require a lower voltage, we will use an LM7805 to regulate the voltage to 5V. Tracking the level in batteries is important to ensure they don't run out during a run.

1.2: Functional Requirements

The Micromouse is required to be able to solve an 8x8 maze from the start (bottom left) to the finish (top right). To do this it must always maintain awareness of its position in the

maze, which we plan to complete this by using maze mapping algorithms to record the path for an optimal route, to do this it must be mapping its path and have positional information. Must perform 90 and 180-degree turns, move forward and backward a cell. To be able to navigate the maze the Micromouse must be able to utilise the sensor data it receives and make the appropriate changes in the input for the stepper motors. Moreover, the Micromouse must be able to understand if it is in a corridor or at a junction and if so then what type.

Chapter 2: Choice of Sensors and Sensor logic

2.1 Sensor Types, Logic and Position

On the Micromouse we will have three distance sensors, on the front and wings of the robot so that we can record the distance between the walls of the maze and readjust our position if there are any issues. In addition to this we will also have 7 IR sensors, four of them will be on the wings of the Micromouse to detect the walls. The other IR sensors will be located at different angles at the front so that we can work out what kind of junction the Micromouse is at. The IR sensors will be shielded to reduce cross signal detection for neighbouring sensors. For the IR sensor logic if a left wall is not detected the speed of the right motor will be adjusted and vice versa, if both walls are detected by the sensors in the wings, we are centred, front wall detected prepare a 90-degree turn.

2.2 Solution to Noise and Interference Problems

The solution to reduce noise and interference problems we may have when the Micromouse is in operation is to use an RC low-pass filter to remove any spikes in the real-world data we get as if we have poor information collected by our sensors, the Micromouse will take unwanted actions and will fail to reach the goal.

Other solutions include using components such as capacitors to stabilise the supply voltage and reduce power supply noise. Furthermore, using components such as diodes to prevent voltage spikes and protect other components. Lastly, we can also use star grounding layout to reduce noise from the motor that will affect sensor readings.

2.3 Calibration Methods

Calibration is important so that we know how accurate the readings the Micromouse gives is. Methods to check this include check all components like motors and IR sensors give the same readings when experiencing the same input. Furthermore, checking the IR sensors minimum and maximum range as well as testing against a variety of different wall distances. In addition to this, motors need to be checked that they rotate the same speed according to the input. Lastly, trial runs to see if any sensor needs to be recalibrated or replaced.

Chapter 3: Choice of Motors and Steering Mechanism

3.1 Motor Types

For the Micromouse we will only be using stepper motors due their improved accuracy and precision over DC motors. Speed is a secondary requirement to the Micromouse as it doesn't matter how fast we can go if we cannot reach the end, it is essential that the Micromouse can do accurate 90° turns.

3.2 Steering Method

The steering method we have opted for is the turning on the spot which involves one wheel rotating forward and the other rotating backwards. This is ideal as it is easier to make 90° turns, easier to control and is suitable for mazes with similar size cells.

3.3 Motor Driver Selection

The motor driver we are using is the ULN2003 because of the current it can handle, it has its own diodes for motor protection, and it is easy to implement into a circuit. It is a motor driver most ideal for unipolar stepper motor.

3.4 Dimensions of Micro Mouse

The Micromouse and all of its components must fit within a 12cm x 12cm footprint, excluding the sensor “wings” which are permitted to extend over the tops of the maze walls. This size constraint ensures the robot can comfortably navigate a maze where each cell measures at least 16 cm × 16 cm, leaving adequate clearance for turning and alignment.

The Micromouse will have a minimum height of 8 cm. The sensor wings will be positioned at approximately 6 cm above the ground, allowing them to reliably detect the maze walls, which are 5cm tall. This ensures the sensors maintain a clear line of sight over the wall height while remaining structurally stable.

Chapter 4: Electrical Power Requirements and Power Distribution

4.1 Current and Voltage Requirements

Table 2: Voltage Requirements

Component	Voltage
Y129 Unipolar Stepper Motor	12 Volts
L297 Stepper Motor Controller	5 Volts
OM11043 LPC1768 MBED Board (internally runs at 3.3 volts)	5 Volts

ULN2003 Motor Driver Chip	12 Volts
LM7805 Voltage Regulator	5 Volts
IS471 IR Receiver	5 Volts
OP165D IR Emitter	1.2 Volts

Table 3: Current Requirements

Component	Current Consumption
Y129 Unipolar Stepper Motor	160 mA per coil
L297 Stepper Motor Controller	20 – 40 mA
ULN2003 Motor Driver	1 - 2 mA per input
OP165D IR Emitter	20 – 100 mA
IS471 IR Receiver	5 - 15 mA
OM11043 LPC1768 MBED Board	30 - 50 mA
Ultrasonic Sensor (HC-SR04)	15 mA -20 mA

4.2 Proposed Battery Technology

The Micromouse will be powered by eights AA batteries, each with a nominal capacity of approximately 2250 mAh. With enough energy for prolonged testing and competition-length operation, this setup offers a readily available and reasonably priced power source.

The batteries will be kept in an 8-cell AA battery holder. The holder fits snugly within the chassis arrangement thanks to its approximate measurements of 57-63 mm for width, and 28-30 mm for heights. To keep it small and easily accessible, it will be installed on the surface of the chassis, just below the Veroboard.

The 75g holder and 8 AA batteries, each weighing 23g, make up the battery system's total mass of 259g. Its location is crucial because this is one of the robot's heavier subsystems. The micro mouse's balance, traction, and manoeuvrability could all be adversely affected by improper placement. To counteract this, the battery pack will be placed as low and in the centre of the chassis as feasible to ensure stability during quick turns, acceleration and deceleration



Figure 1. Multicomp Pro MP000320 8-cell AA snap-on battery holder (Multicomp Pro 2024).

4.3 Heatsinks

Heatsinks are required in the Micromouse design to ensure safe thermal performance, particularly for the voltage regulator. When stepping the supply voltage down from 12V to 5V, the regulator must dissipate a significant amount of power as heat, and without adequate cooling it may operate at temperatures that reduce efficiency or reliability. A heatsink is therefore necessary to maintain stable operation.

Other components, such as the motor driver and sensor circuitry, do not require heatsinks. These devices operate at lower current levels and are mounted in open airflow within the chassis, allowing heat to dissipate naturally. As a result, passive cooling is sufficient, and additional thermal management is unnecessary for these parts.

4.4 Voltage Regulation

To regulate the supply voltage within the Micromouse, a 7805 linear voltage regulator is used to step the input voltage down from 12 V to a stable 5 V. This ensures that all 5 V-dependent components receive a consistent and safe supply, protecting sensitive circuitry from voltage fluctuations or surges. By providing a regulated output, the 7805 maintains reliable operation across the microcontroller, sensors, and supporting electronics.

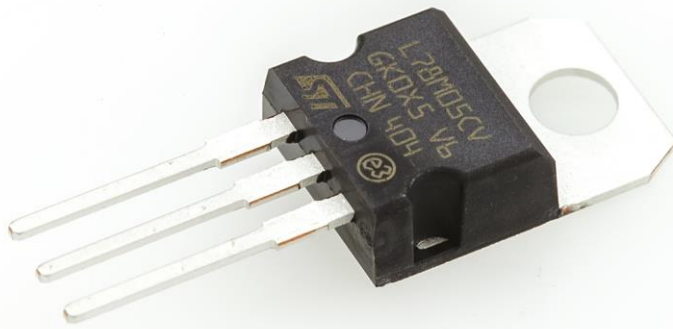


Figure 2. 7805 linear voltage regulators used in the Micromouse (Texas Instruments 2016).

Chapter 5: Proposed Chassis Design

5.1 Proposed Layout and dimensions of chassis.

The chassis, excluding the wings, will fit within a 15cm * 15cm footprint. It will be U-shaped and be about 8cm tall. With two arms extending 2cm – 3cm outward on either side to interface with the wings, the core body, which houses all internal components, will be about 10cm broad.

Internally, the stepper motors are positioned on the chassis, and their shafts connect to the wheels, which are situated beneath the U-shaped overhanging arms through specially designed holes. The remaining electronic parts are positioned on the same horizontal plane as the overhanging arms, where the batteries are in the centre section. Since the chassis offers the most convenient and accessible surface for component integration.

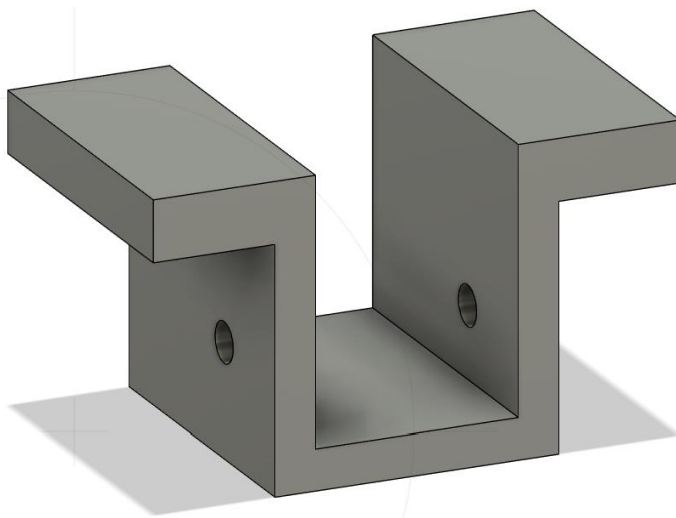


Figure 3. Proposed Chassis Design

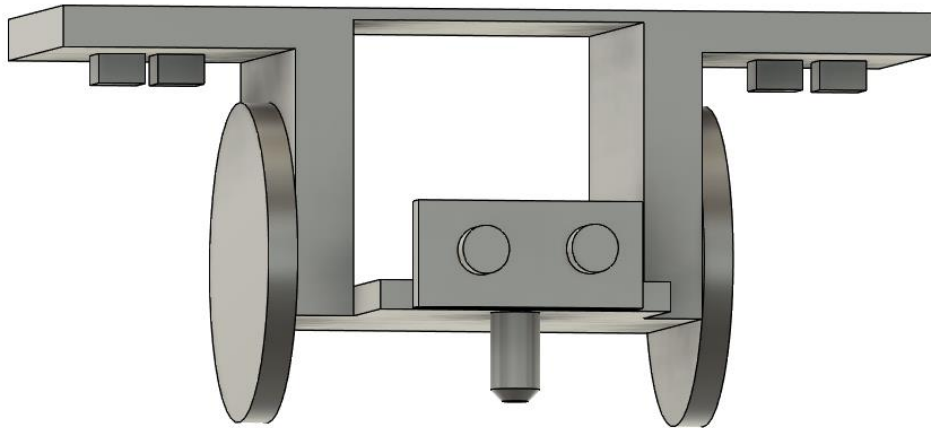


Figure 4. Proposed Chassis Version 2 Design with Wing

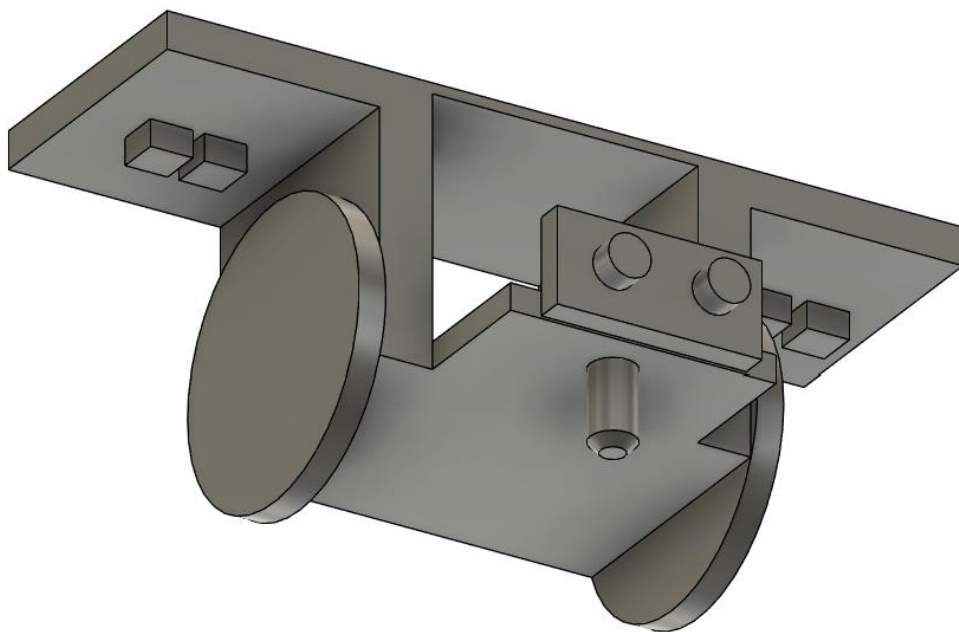


Figure 5. Proposed Chassis Version 2 Design with Wing

5.2 Circuit, Wheel and Motor Mounting Method

The circuitry will be positioned on the upper surface of the chassis and secured in place using combination of spacers and screws. The use of spacers is deliberate – they elevate the board slightly above the chassis surface, which not only protects the underside of the board from potential short circuits but also makes it significantly easier to access, remove, and reinstall the board whenever repairs or modifications are required. The wheels will be mounted directly onto the shafts of the stepper motors, ensuring a tight, direct coupling

between the two components. They will be firmly fastened using screws to eliminate any looseness or play connection. This is a critical design consideration, as the Micromouse to drift or move unpredictably rather than following a precise controlled path. To anchor the stepper motors themselves to the chassis, a combination of screws and mounting brackets will be used, ensuring the motors remain rigidly fixed in their intended positions even under the vibrations and forces generated during operation.

5.3 Design for Maintainability

The layout of the Micromouse is designed so that it is easy to access components to replace or repair them if a screwdriver as components will be connected to the chassis by screws. The wiring used will be colour coded depending on what it is connected to, so it is easier to understand what is connected to what and it's quicker to make fixes. The wiring to components that aren't soldered will be jumper wires so it easier to change them out. For example, the stepper motor will connect to the pin holder that are soldered so it is easy to remove the stepper motor to replace it is necessary.

5.4 Sustainable Design Aspects

Sustainability is an important consideration in the design of the Micromouse chassis and wheels. Materials such as aluminium offer several advantages, including being lightweight, strong, and highly recyclable. Aluminium can be reused repeatedly without significant degradation, making it a suitable choice for reducing environmental impact.

However, the sustainability of the overall design also depends on factors such as ease of disassembly, the recyclability of electronic components, and the durability of the chosen materials. While aluminium is recyclable, other components-such as PCBs, sensors, and wiring-are more difficult to recycle and may require specialised processes. Ensuring that materials are durable and long-lasting reduces the need for replacements, minimising waste over the robot's lifetime.

Table 4: Material Comparison

Material	Advantages	Disadvantages	Recyclability
Aluminium	Lightweight, strong, corrosion-resistant, long lifespan	Higher cost than plastics; requires more energy to manufacture	Excellent – can be recycled indefinitely
PLA (Bioplastic)	Biodegradable, easy to 3D-print, low cost	Brittle, low strength, degrades under heat	Good – industrial composting or recycling
ABS Plastic	Stronger than PLA, durable, impact-resistant	Petroleum-based, not biodegraded	Moderate – recyclable but less commonly processed

Steel	Very strong, extremely durable	Heavy, not ideal for small robots	Excellent – widely recycled
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Chapter 6: Broader Design Proposals

6.1 Modes of Operation

A Micromouse robot can navigate a maze using several different algorithms, each with its own advantages and limitations. One of the simplest approaches is wall following, such as the left-hand rule, where the robot continuously maintains contact with the left wall regardless of the junctions encountered. Although easy to implement, this method is slow, cannot guarantee the shortest path, and may fail in mazes that are not fully connected (**IEEE Robotics and Automation Society, n.d.**). [3]

A more efficient and widely adopted strategy is the flood-fill algorithm, which assigns a numerical distance value to every cell in the maze based on its estimated distance from the goal. As the robot explores, these values are updated dynamically, allowing the Micromouse to move toward neighbouring cells with the lowest cost. This results in a more optimal route and significantly faster completion times once the maze has been mapped (**Otten, 2011; Siegwart, Nourbakhsh and Scaramuzza, 2011**). [4]

In addition to competitive navigation modes, the Micromouse can operate in specialised practice modes. For example, a tunnel run mode allows the robot to test its sensors, encoders, and motor control in long straight corridors without the complexity of junctions. These modes are useful for calibration, debugging, and validating subsystem performance before full maze-solving runs (**IEEE Robotics and Automation Society, n.d.**). [5]

6.2 Algorithms

Flood fill algorithm will be the most efficient to use. This is carried out by assigning each cell a cost to our goal which will have the value of 0. The cells next to the goal get the value of 1 and the neighbours get 2 and so on. From the start of the maze, we will move to the adjacent cell with the lowest value. By doing so it guarantees that it will eventually find the shortest path. At the start the mouse assumes there are no walls and all the paths are open. The mouse will check front, left and right, if a wall is detected with the sensors, it updates the maze map, if a wall block what it thought was open it recalculates and continues to move to the lowest neighbour. This ensures the algorithm adapts constantly. (A. Wu, 2025) [6]

References

- [1] Danie Conradie, "The Fascinating Evolution Of Micromouse," *Hackaday*, May 29, 2023. <https://hackaday.com/2023/05/28/the-fascinating-evolution-of-micromouse/>
- [2] Farnell, "MP000320 BATTERY HOLDER, AA, SNAP ON," *Farnell UK*. <https://uk.farnell.com/multicomp-pro/mp000320/battery-holder-aa-snap-on/dp/3126554>
- [3] S. Wan, "Micromouse Competition Rules," 2025. <https://ieeer1.org/wp-content/uploads/2025/01/Micromouse-Competition-Rules.pdf>
- [4] P. Harrison, "Minos 2011 Report - Micromouse Online," *Micromouse Online*, Apr. 13, 2011. <https://micromouseonline.com/2011/04/13/minos-2011-report/>
- [5] R. Siegwart, Illah Reza Nourbakhsh, and Davide Scaramuzza, *Introduction to autonomous mobile robots*. Cambridge, Massachusetts: Mit Press, Piscataway, New Jersey, 2011. <https://mitpress.mit.edu/9780262015356/introduction-to-autonomous-mobile-robots/>
- [6] A. Wu, "Micromouse," *Uci.edu*, 2025. <https://ieee.ics.uci.edu/micromouse/floodfill.html>